Assigment One

Section A

Database

① Database is a structure simply made of columns and rows, that allows structured data (information) to be accessed, changed / modified, managed controlled and updated.

(1) Database is essential because

* **Managment**

1.1 managent

Data can be easily managed

* **Accuracy**

When structured data is stored in a database accuracy is normally there

* **Updating**

changing and corrections are also easily done when data is in a database

* **Security**

when data is stored in a database it is not easily accessible as an spreadsheet group or individuals, can setup secure security Passwords so that it cannot be easily accessed

Real world example

* Online banking : e.g Capitec APP
* Streaming Services : e.g Showmax

Data base

Name	ID No	Balance
Noma	5708110276	R500
Zadua	7508112511011	R5000
Phila	5706125904085	R50000

Assignment One

② A Datawarehouse

- Is a special system used to pull data from many diffent places, like sales software e.g (POS) Point of Sale system from the stores or online platform
- It stores it all in the warehouse sytem
- It doesnt store data in real time but later to avoid the system to crush or analyze later

- Difference between Database and Database Warehouse

Database	Database Warehouse
* Real Time	* After Hours
* What happening right now	* What is happening in the future
* Is the Processor right now	* The Thinker on how are things progressing and what does the future hold

So Data Warehouse

- Is the cleaner
- Time Traveler - sees the future
- Seeing the future -
- Is also the to see the bigger future

Assignmen One

(3) Datawarehouse VS Datalake

- A Data warehouse is a structured, optimized repository for pre-processed data used for (BI)

* Whereas Data lake is flexible; cost effective storage for raw, unstructured and semi structured data.

Key difference	
Data Warehouse	Data lake
* Uses Structure and process Data	* Uses Raw, structured, semi structured, unstructured Data
* Cost is Normally higher	* It is Cost effective / or lower Cost
* Its purpose is Reporting, BI; operational analytics	* Purpose is Machine learning, AI, Exploratation.

Dart Mart What is it

(4) It is a specialized, smaller subset of data warehouse designed to serve a specific analytical needs of a single department, such as a marketing finance; or s Sales.

How does it relate to Datawarehouse

They Both have the same purpose and the purpose is to CRUD

Create READ Update Delete

- the only difference is that Datawarehouse does it in a bigger or large Scale

yet Data mart does in smaller scale specific departments.

Assignment One

(5) Difference between

Structured Data	Semi Structured Data	Unstructured Data
• Its organised Data • Organized in a form of Rows and Colomns	• Semi Organized Data • It used tags/meta data for hierarchy	• Unorganized Data • non set and difficult to analyze Raw Native format
Examples table and rows	Examples JSON, XML, Avro Parquet, ORC	Example Photos, videos, music, email, PDF

	Name	Surname	ID
1	Nama	Njoloo	123
2	Zedua	Zuma	045

(b) Data Modelling

* It is the process of creating a visual representation of the entire information system of parts of it to communicate connections between data points and structure

Importance

* It is important because when designing a system without a data modelling / model is like building a hotel without architectural plan.
  = All these u Play a huge role in Data Modelling

- Standardization
- Perfomance Optimization
- Data intergrity
- Cost Reduction

2 TYPES OF Data Models

Conceptual Data Model	Physical Data Model
This ensures that the concept allignes with the bussiness goal. It focuses on what data will the system contains and not focus on the technicality	This focus on the technicality and how data will be stored.

Assignment One

a) Data Mining

It is the automated process of analyzing massive datasets so that it identifies hidden patterns, trends; and correlations that are not obvious to us as humans.

How is it used in business?

- **Predictive Analytics:** It predicts the future on business as to the product demants, trends or potential equipment failure.

- **Risk Management:** This detect abnomalities in financial transactions to prevent fraud.

- **Customer Segmentation:** this identifies a group of customers with similar purchasing habits to create targeted marketing campains.

Examples:

TakeALOT	Makro / Game Store
Uses data mining to analyze the history when customers are browsing past purchases, when this happens you can see the data mining in action.	Makro or Game store (POS) Systems tracks exactly which products are sold the most and in which locations. In this case the store are able to maximise the stock in the area and increase revenue

Assigment One

① (DBMS) Data management System

It a bit of software that manages databases. It ensures that information is stored securely, retrieved, and modified accurately.

Popula DBMS Platform

① MySQL : It an open-source relation DBMS (RDBMS) that uses structured Query language (SQL)

Used For : Speed, realibility and ease

Content management systems like WordPress & eCommerce Platforms, where reliable, read-heavy Performence is required

② PostgreSQL : This is a highly advanced object-relational database system known for it distict adherence to SQL standards, extensibility, and support for complex data support.

Used for : It is ideal for systems requiring high precision, such as financial transaction Processing

③ Oracle Database : This was developed by Oracle Corp. It is a commercial model (RDBMS). It is designed for corparates/enterpice-level applications.

Used for : Used to manang corpate data or massive data access global operation such as human resources and supply Chain management.

Assiament One

(8) (4) **MongoDB** this is a NoSQL database
it stores data in flexible, JSON-like documents rather than traditional tables

Used for : This is excellent for applications where data structures evolve rapidly such as user profiles management in social media apps (facebook) or even storing and handling unstructured data.

(9) **ETL** : Exctract, Transform and LOAD
this is a stage where data is been used to move raw data into a staging area (DW), enable transform without interruption on production, then upload without data loss or errors.

Extract : The reason for this is to exctract /retrive data from multiple source without disruption

Transform : This is to transform (change) raw data into something / into meaningful format

Load : This is to make that ~~that~~ been transformed "Loaded" into a final step, so that it is ready for consumption.

Assigment One

(10) Formal Data Used in Analytics

① CSV (Common Separated Value)

This is a simple, plain text format ~~text~~ format where each line represents a row or a column.

NB Not to use for larger Scale analytics.

② JSON (JavaScript Object Notation)

This is a light weight text based format that stores data in a key-value pairs. It supports hierachical and nested structure. making it more flexible than CSV.

NB not to use for larger scale analytics

③ Parquet (Apache Parcequet)

This is a binary columnar storage format. Instead of staring data row by row it groups data by colomns. this allows extreme compression and effeient retrieval.

NB Avoid when you need to write data record by record in real time or frequently updated individual rows, as columnar formats are not optomized for frequent

Use : Analytical Queries

④ Avro : (Apache Avro)

A binary row based serialization format that stores its schema (metadata) in a JSON format within the file itself.

NB : You need to write data record by record in real time or frequently update individual rows, as a columnar formats are not optomized for frequent

Assignment One

10 CSV	JSON
When to use This is Ideal for simple data exchange	When to use This is best for lightweight, text based format that stores data in a key-value pairs
Parquet	**Avro**
When to Use This is used for analytical queries	When to Use This is perfect for streaming data pipelines

(11) AI (Artificial Intelligence) Section B

Its a technology that enables computers and machines to simulate human learning comprehension, problem solving, decision making, creativity and autonomy.

i) Narrow AI (WEAK AI)

Narrow specifically designed to master one specific Job, It operates under limited constraints

ii) General AI (Strong AI)

This can master tasks, learn and apply intelligence to any problem just like humans!

Assigment One

(e) ML (Machine Leanin)
Machine learning is a Subset of AI concerned with training modules to allow computer to mimic humans thoughts and decesion making without explicit Programming

Superuised learning (learning under Label Data)

Unsuperuised learning (learning under Unlabeled

Reinforced learning (learning unde punnishment responce and reward lea)

ML (Machine learning) VS Traditional Programming	
This is when a branch of AI that allows computer to figure out or learn patter without being Programmed	Yet Traditional Programming need human to give input and rules to get things done

Assignmen One

13 Deep learning

DL is subset of ML and AI
DL takes time to gather data but quit quick to generat answers
- So it learns deeply about data compared to ML
- And it differs according to efficiency due to the amount of data it includes

Relation between ML and DL in AI

AI	ML	DL
Uses deep Learning (DL) models to create new, original content	Analyse data to make Predictions	Uses multi-layered netrual networks for complex pattern recognition
Part of AI	Parts of AI	Part of AI
This generate/ ~~gather~~ data to generate Answers	This also gather data to generate Answers	Also gather data to generate Answers

Assiament One

(1) Natural language Processing (NLP) ?
What is it ?

① This is all kind of AI branches that teaches the computer or systems to read, interpret and respond to human language.

3 Real World applications of NLP

(1) Mc Donald Kioski : when everytime a human Puts an order on the system or kioski that info is taken or data is been translated to the back for someone to understand

But before it goes back to someone or human to understand system has to code, digest, read then respond again into human language.

(2) Language translation (Siri) : this listens to human voice into the system then translat to the system to get answers, or perform the request.

(3). Custome Suppat business : (chabots / Tobi) these are mostly used by business to answer queries 24/7 for customer when a human is not there.

Assignment One

(15) **LLM** (Large Language Model)

* This is a type of AI that is trai
to take huge amounts of text so tha
it can understand human language

LLM	Standard ML
* Does Large task e.g handle many languages at one, such as summarizing, writting and Answering	* Does Specific Task e.g spam email detector

2 LLM examples?

* Google Gemini and GPT

The analy human text or generate text analy data, write code all this generated from a simple form as text analyzed then transferme into comple or constructed data for consumption.

(16) **Generative AI**

Is a Subset of AI that create new content including Text images, audio and code by identifying patterns in existing data.

(1) At High level it uses (LLM) Large Language Models or diffusion models to determine the most likely element in a sequence

Assignment 1

16 (ii) 3 Generative AI Tools

- **Midjourney**: This is an image generator that descriptive text prompts and transform them into high quality, artistic visual graphics or photostatic image by diffusing noise into structured pixel.

- **ChatGPT**: This is a conversational assistant that processes natural language to answer complex question
  i.e it answers essays; summaries docs and helps with programming task by predicting contexually relevant text

- **GitHub Copilot**: This is an AI - powered coding companion that lives within a developers code editor, suggesting entire lines or blocks of codes in real-time to accelerate software development.

Assiament 1

(17) Traning Dataset?

① It a collection of Informatio
Information such as text, Images or spreadshee
This is used to teach AI model to recogni
patterns & to make predictions.

Critical because?

- Quality is critical because model accuracy depends entirely on the data consumed.

② So it is baised on the accurary or quality of data produce, so that the AI may produce accurate results.

What can go wrong?

③ If the data is baised or incorrect AI informatio producess will not be accurated or false

(18) Computer Vision

It is a technology that machines uses to automatically recognize images and describe them accurately and effecently.

2 real-world cases of computer visio.

- Foot track in Retail stores
- Facial biomatric to unlock phone.

Assignment 1 Section C

(19) **API ?** (Application Programming interface)

Its a set of rules and protocols that allows different Sofware application to communicate and share data to one another.

Real World analogy: (MTN Store)

The Customer (me) I am the "client" application I know what I want but Cannot access to the safe to get my phone i want

The Back office (The server) Server database) It contains the device and accessories I need, but I dont have access to the back office

The Catalogue : (THE API document) This list every device and add on accessories available and information i need to provide to get this.

The Sales Consultant : (API)
I give out my order list she goes to the back office ensure that includes accessorie my order for the device and accessories are ready then brings it to me. (data repo

Assiameng 1

(20) **API KEY** (Application Programming Interface)
These are generated strings of characters used to authenticate and authorize a specific application or project when it is requested for / to a service.

Purpose API key
A primary purpose for the API key is to act as a Identifier for software application

How should it be store and protected
They must be store and saved with caution as a password.

Consequences of Public exposure
Can lead to severe security and financial issues.

(21) **Rest API**
Its a software used for design network application that allows two computers system to communicate over the internet.

REST (Representational State Transfer)
Main HTTP methods

GET: Used to get or retrive data from the server

Post: Used to Create a new resours on the server

Delete: Used to remove a specific resource from server
Completely replace an existing

Assigment 1

22 JSON (JAVAScript Object Notation)?

It is a lightweight text based format commonly used for storing and exchanging data.

Commonly Used when working with APIs & exchang

Reasons

* That it is lightweight and fast
* Language Agnostic
* Human Readable
* Easy to Parse

Example of a JSON object

It is a way to organize data using keys and values

```
{
"Name": "Noma",
"Id": 011,
"isStudent": false,
"hobbies": ["soccer", "movies"]
}
```

23 What Prompt Engeneering?

It is the art of crafting Precise instructions to guide AI Tools towards the best possible results.

Importance: It also a vital skill to master AI models like Claude, ChatGPT if this is mastered you can avoid errers and save time with errors

Assigment One

23 2 Example of Poorly written prompts and an improve version of each

Poor vs Improved

EXAMPLE 1 Creating a diet meal Plan	
Poor Prompt	Improved Prompt
Why is it poor: It lack content about diet plans goal and what food should be included leading to generic responce	A dietian, creates a month plan with the meals that should be included and when and how it should be taken. and provides this in a table format

Content creator Creating a Blog on Social

EXAMPLE 2 Poor Prompt	Improved Prompt
Why is it poor: the content written lacks writing skills It is not structurred and doesn't include subheads and conclusion	Provides blog post, content is structurred and includes subheads and conclusion, there for it give out positive responce and productive

Assigment One

24 Map function.

This a powerful tool in Programming and data processing

- It is used for transforming and collecting Items into a new collection of the same size.

Python

```
number = [2,3,4,5]

ii Applying a Lambda (anymous)
doubled = list(map(lamda
x: x*
Print(doubled) # Output
[4,6,8,10]
```

take fuctions first then the list

JavaScript

```
Const numbers =
[2,3,4,5];

//using the arrow function to do
Const doubled = numbers.map(num
Console.log(doubled);
// . Output
```

directly on array

Imagen this a you had a list of numbers and want to creat a new list where every number is doubled. And thathats the example.

Assiament One

What is Cloud Computing ?

Its a delivery of computing service, that can not be touched by human hands this includes storage, servers, software - over internet, database doesn't rely and hardware like usb and hard drive.

3 main Cloud service models

IaaS (Rentool infrastructure)
e.g Icloud, google drive ect.
This gives users the control over their IT resources without having to maintain physical hardware.
(AWS) (Amazon) ect.

PaaS (development infrrastructure) e.g Google App engine
This gives a framework and tools that developers use to build, test and deploy applications

SaaS (Ready to use software) e.g Microsoft 365
This delivers fully functionetional software applications over the internet on subscription basis.

Assigment One

2b Version Control System (VCS)

This is a tool that manages and tracks changes to files over time, acting like a time travellar machine for your projects.

Git : It is a popular version control system it tracks changes in files.

The reason it is used by data analysts and developer they can collaborate with other data analyst or developers and for is safety

- **Commit** : take is as a submit, save or checkout points in your project

- **Branch** : Parallel universe for the code This allows you to step away from the main stable version of a project to work on the new feature or experiment in privat

Assignment ONE

(89) Jupyter Notebook

This is an open-source web application
It is used to create and share documents
that combines live code, equations
computation outputs and narative text.

(i) Commonly used with Python

(ii) Popularity among Data analysts and Scientists

Reasons

(i) Executes code interactively

(ii) Tell a Story wit data

(iii) View inline visualizations

(iv) Cache data in memory

(v) Ensure reproducibility

Assignment One

(a) What is Python

It is a high level interpreted programming Language. designed to make thing easy and reliable.

Importance of Python

It has a Massive Community support
Seamless Intergration
Vast library Ecosystem
English - like Syntax

Four libraries for Data work

① **Pandas** : Also know as the bread and buta of data analysis
It is used to handle tabular data.
(like spreedsheet)
I allows you to clean, filter and aggregate large datasets effeciently using its structure called a "Dataframe"

② **Numpy** : mostly used for numerical structure data, calculations and handles large multi-dimensional arrays

③ **Matpolotlib** : Use for Visualization
It allows users to turn numbers into clear customized visual reports. and this is done with graphs, bar charts and histograms.

Assigment One

(4) Essential Python Libraries for Data work

(4) Scikit-Learn :

This is an essential library for machine learning (ML)

It provides pre-built tools for common AI task

Assiament One

Section D : Data ethics, Governance and Governance

29

Data Privacy : This is a right to privacy to be able to control personal information such as identity, names, address and online behaviour is collected and used or shared by organisations

important because privacy is about governance of personal information.

- Protection Against fraud
- Prevents Exploitation
- Build Trust
- Personal liberty

GDPR : Is the EU's new data protection Law It is a set of strict rules and regulations implemented to replace outdated laws effective as from May 25, 2018

Countries it applies in are EU/EEA and organazation targetting them.

Assiament One

30 **Data Ethics** : Is a branch ~~of~~ applied ethics that addresses the moral obligations and guidelines Principles surrounding the collection, storage, analysis and usage of data.

Example of Unethical use of data or AI

① Algorithmic Bias in hiring tools

Scenario : when a company uses AI system that screens resumes but the AI is trained on the historic data from a male dominated industry

Unethical : this will be unethical because the system reads from the previous employment history and that only males or 90% of male where employed

Assigment One

(1) **Data governance**: is the strategic orchastration of people, processes and technologies to ensure that an organisation's data is accurate, secure, and usable throughout its lifecycle.

key Components of Data Governance Framework

Data Stewardship: The appoiments of individuals (Data Stewards) responsible for the day-to-day management

Data Quality management: Standardized procedure to monitor clean, and validate data

Complience and Security: Right controls to proted sensitive information and ensures the organisation adherers to the legal requirements like GDPR and HIPAA

Data Architecture: The underlying infrastructure and tools that support data flow.

Assignment One

32 Bias in AI

This occurs when algorithms produce prejudiced results due to flawes data and design choices

AI Bias happens when a system in AI pulls up systemetical favors or discrimination againts certain groups after mimoning mimicing the historica behaivors, (like race).

example

The Cause / Scenario : When a compony uses AI with the hiring Process then the system read the historical info that only men or certain race were employed by the Company therefore it Shall choose from the historical Data

The Result : Will be then based on race and men based

Consiquence or Results
When AI Bias is left unchecked then is east to get a lot of errors and severe negative output and long lasting as well.

Assiament One

TABLE Explaining Betwee all 4 Roles

Role	Description	Key Skill	Core Tool	Primary Focus
DATA Engineer	The Architect of data world They focus on the collection of data.	ETL Database Design Big Data	SQL Apache Spark Airflow AWS / GCP	Data Pipeline & Architecture
Data Analyst	The storyteller They examine Structurred data to identify Trends	Data Visualisation SQL Statistics	TABLEU Power BI Excel, SQL	Descriptive Analytics & Reporting
Data Scientist	The Researcher They use advanced Statistics and machine learning to predict future trends.	Machine Learning Math Python / R	Python R, Scikit-Learn Jupyter	Predictive Modelling & Research
AI Engineer	The builde of Intelligence They focus on creating systems that can perform task requiring Human-like intelligenat	Deep Learning, NLP, Computer Vision	Tensorflow PyTorch Kerras OpenAI. API	Neural Networks & Model Deployment

Assigment One

(34) **BI (Business Intelligence)**

Its a process of collecting, analyzing and transformin raw data into meaningful insights to guide strategic decision

- **Microsoft Power BI :**
A platform tha intergrate deeply with microsoft services (lik Excel and Azure) to create interactive dashboard and reports

- **Tableau :** know for it powerful visual analytics it allows users to explore data through intuitive drag-and-drop interfaces to uncover hidden pattern.

Assignment One

Section E

Personal Reflection

35 Concept I found from the assigment most interesting and suprising

(1) It was interesting just learning about the fundementals of Data suprising was when I learned more about the Data world and all the terminology they use, previously that these words used in data are just words used for programming but not

36 Most difficult to learn was to describe each terminallogy Word, define translate in my own word and describ

37 Data and AI are growing at a fast pase in the world to a moment that I think every person currently working will be replaced by AI in the future

Resource used

(1) I used google 90% of the time Some answer was from microsoft, and Google Search